Chain Rule: Derivatives of Composite Functions

Calculus Worksheet · Grade 11-12

Name: ______________________________ Date: _______________ Score: / 10

Learning Objectives

- Identify composite functions of the form f(g(x))
- Apply the chain rule to differentiate functions raised to a power
- Simplify derivatives obtained using the chain rule

Find the derivative of each composite function using the chain rule and simplify your answer.

1. Find the derivative using the chain rule.

$a(x) = (5x+1)^3$

Answer: ______________________________

2. Find the derivative using the chain rule.

$m(x) = (3x-2)^7$

Answer: ______________________________

3. Find the derivative using the chain rule.

$b(x) = \sqrt{x+1}$

Answer: ______________________________

4. Find the derivative using the chain rule.

$c(x) = (4x)^{\frac{2}{3}}$

Answer: ______________________________

5. Find the derivative using the chain rule.

$h(x) = (2x^2+3)^5$

Answer: ______________________________

6. Find the derivative using the chain rule.

$p(x) = \sqrt{3x^2-5}$

Answer: ______________________________

7. Find the derivative using the chain rule.

$q(x) = (x^3+2x)^4$

Answer: ______________________________

8. Find the derivative using the chain rule.

$r(x) = (7x + 4)^{10}$

Answer: ____

9. Find the derivative using the chain rule.

$s(x) = \sqrt[3]{6x - 1}$

Answer: ____

10. Find the derivative using the chain rule.

$t(x) = (x^2 - 4x + 1)^6$

Answer: ____

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ANSWER KEY & SOLUTIONS

Encourage students to clearly identify the outer function f(x) and inner function g(x) before differentiating.

Solutions

1. Find the derivative using the chain rule.

$a(x) = (5x + 1)^3$

→ *Identify the outer function as the parentheses cubed and the inner function as 5x plus 1.*
→ *Differentiate the outer function to get 3 times the parentheses squared.*
→ *Multiply by the derivative of the inner function, which is 5.*
→ *Combine the constants 3 and 5 to get 15 times the quantity 5x plus 1 squared.*

Answer: $a'(x) = 15(5x + 1)^2$

2. Find the derivative using the chain rule.

$m(x) = (3x - 2)^7$

→ *Identify the outer function as the parentheses to the seventh power and the inner function as 3x minus 2.*
→ *Differentiate the outer function to get 7 times the parentheses to the sixth power.*
→ *Multiply by the derivative of the inner function, which is 3.*
→ *Combine the constants 7 and 3 to get 21 times the quantity 3x minus 2 to the sixth power.*

Answer: $m'(x) = 21(3x - 2)^6$

3. Find the derivative using the chain rule.

$b(x) = \sqrt{x + 1}$

→ *Rewrite the square root as the inside function raised to the one-half power.*
→ *Differentiate the outer function using the power rule to get one-half times the parentheses to the negative one-half power.*
→ *Multiply by the derivative of the inner function x plus 1, which is 1.*
→ *Rewrite the negative exponent as a fraction with the square root in the denominator.*

Answer: $b'(x) = \dfrac{1}{2\sqrt{x + 1}}$

4. Find the derivative using the chain rule.

$c(x) = (4x)^{\frac{2}{3}}$

→ *Identify the outer function as the parentheses raised to the two-thirds power.*
→ *Differentiate the outer function to get two-thirds times the parentheses raised to the negative one-third power.*
→ *Multiply by the derivative of the inner function 4x, which is 4.*
→ *Combine the constants two-thirds and 4 to get eight-thirds times the quantity 4x to the negative one-third.*

Answer: $c'(x) = \dfrac{8}{3}(4x)^{-\frac{1}{3}}$

5. Find the derivative using the chain rule.

$h(x) = (2x^2 + 3)^5$

→ *Identify the outer function as the parentheses to the fifth power and the inner function as 2x squared plus 3.*
→ *Differentiate the outer function to get 5 times the parentheses to the fourth power.*
→ *Multiply by the derivative of the inner function, which is 4x.*
→ *Combine 5 and 4x to get 20x times the quantity 2x squared plus 3 to the fourth power.*

Answer: $h'(x) = 20x(2x^2 + 3)^4$

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6. Find the derivative using the chain rule.

$p(x) = \sqrt{3x^2 - 5}$

→ *Rewrite the square root as the inside function raised to the one-half power.*
→ *Differentiate the outer function to get one-half times the parentheses to the negative one-half power.*
→ *Multiply by the derivative of the inner function 3x squared minus 5, which is 6x.*
→ *Simplify by canceling one-half with 6x to get 3x divided by the square root of the inside.*

Answer: $p'(x) = \dfrac{3x}{\sqrt{3x^2 - 5}}$

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7. Find the derivative using the chain rule.

$q(x) = (x^3 + 2x)^4$

→ *Identify the outer function as the parentheses to the fourth power and the inner function as x cubed plus 2x.*
→ *Differentiate the outer function to get 4 times the parentheses to the third power.*
→ *Multiply by the derivative of the inner function, which is 3x squared plus 2.*
→ *Write the final answer as the product of these factors.*

Answer: $q'(x) = 4(x^3 + 2x)^3(3x^2 + 2)$

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8. Find the derivative using the chain rule.

$r(x) = (7x + 4)^{10}$

→ *Identify the outer function as the parentheses to the tenth power and the inner function as 7x plus 4.*
→ *Differentiate the outer function to get 10 times the parentheses to the ninth power.*
→ *Multiply by the derivative of the inner function, which is 7.*
→ *Combine the constants 10 and 7 to get 70 times the quantity 7x plus 4 to the ninth power.*

Answer: $r'(x) = 70(7x + 4)^9$

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9. Find the derivative using the chain rule.

$s(x) = \sqrt[3]{6x - 1}$

→ *Rewrite the cube root as the inside function raised to the one-third power.*
→ *Differentiate the outer function to get one-third times the parentheses to the negative two-thirds power.*
→ *Multiply by the derivative of the inner function 6x minus 1, which is 6.*
→ *Simplify by canceling one-third with 6 to get 2 divided by the quantity 6x minus 1 to the two-thirds power.*

Answer: $s'(x) = \dfrac{2}{(6x - 1)^{\frac{2}{3}}}$

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10. Find the derivative using the chain rule.

$t(x) = (x^2 - 4x + 1)^6$

→ *Identify the outer function as the parentheses to the sixth power and the inner function as x squared minus 4x plus 1.*
→ *Differentiate the outer function to get 6 times the parentheses to the fifth power.*
→ *Multiply by the derivative of the inner function, which is 2x minus 4.*
→ *Write the final answer as the product of these factors.*

Answer: $t'(x) = 6(x^2 - 4x + 1)^5(2x - 4)$